

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 3B: The Normal Distribution

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Outline

- ① The normal distribution
- ② Standard normal distribution
- ③ Computing normal probabilities
- ④ More Examples
- ⑤ Outlook

Reading: OpenIntro Statistics 4.1

Normal (Gaussian) distribution

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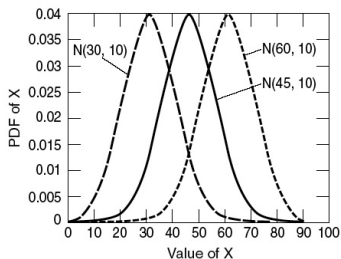
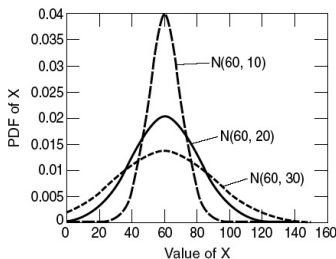
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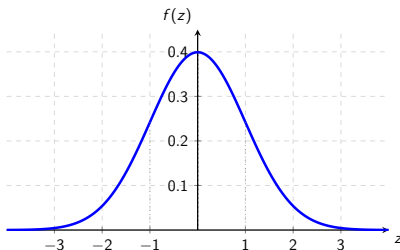


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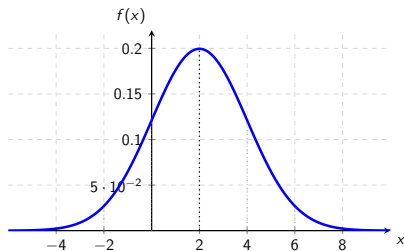
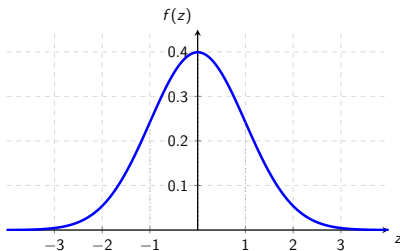


Figure: Standard normal distribution $\mathcal{N}(0, 1)$ **Figure:** Normal distribution with $\mu = 2$, $\sigma = 2$

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where

$$z = \frac{x - \mu}{\sigma} \quad (4)$$

- The standardized normal variable z is often referred to as the Z -score

Standard normal distribution

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The CDF Φ of the standard normal variate Z is given by:

$$\Phi(z) = F_Z(z) \equiv P(Z \leq z) \quad (7)$$

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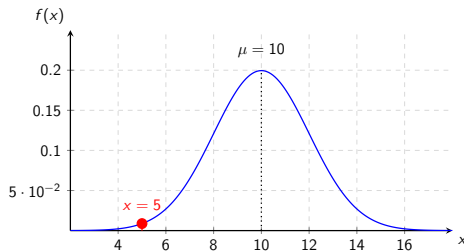


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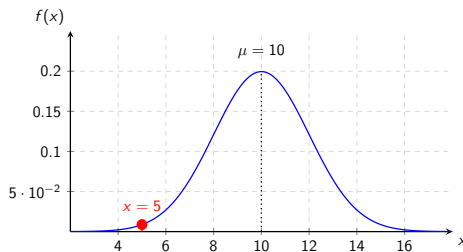


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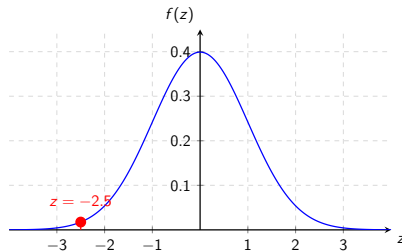


Figure: Standard normal distribution with $z = -2.5$ indicated

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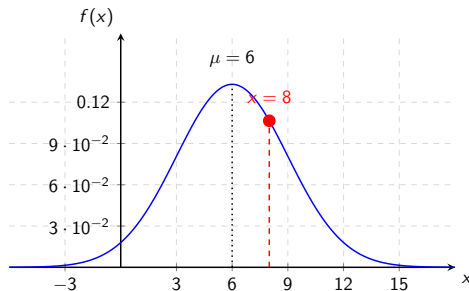


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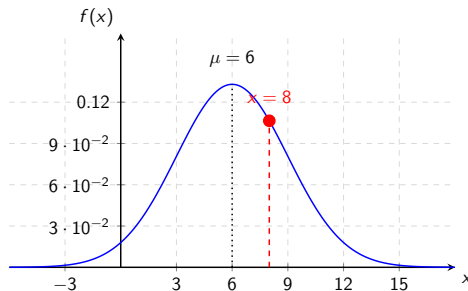


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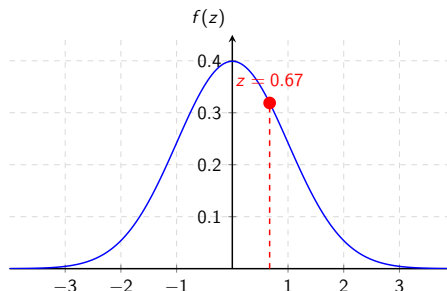


Figure: Standard normal distribution with $z = 0.67$ indicated

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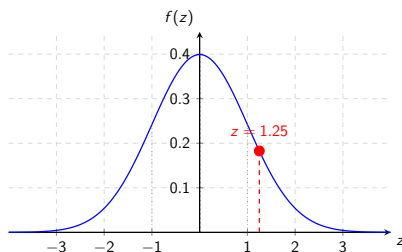


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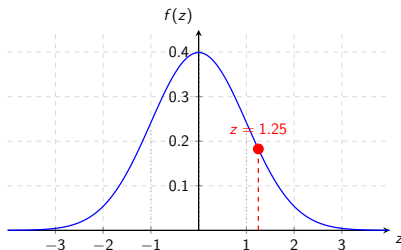


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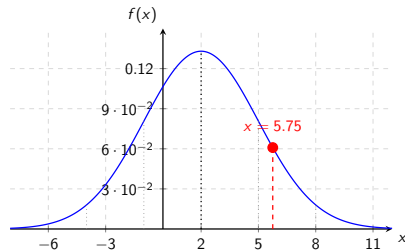


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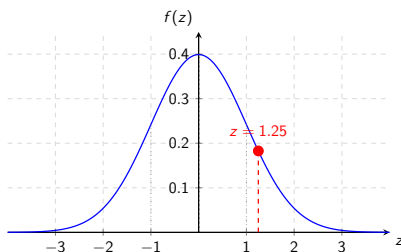


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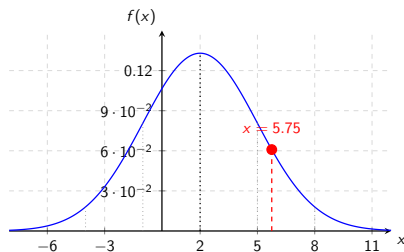


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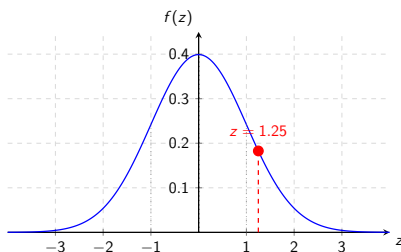


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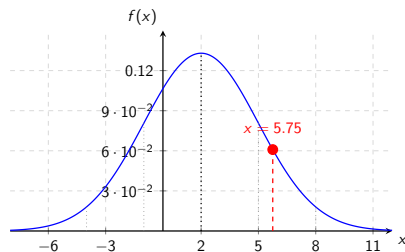


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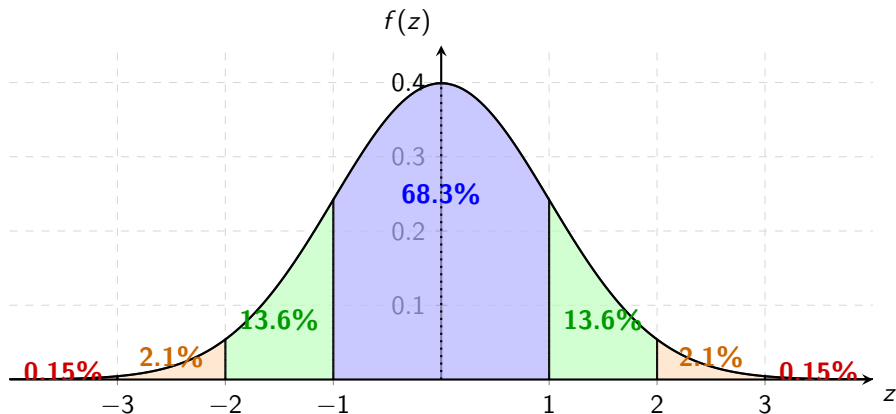


Figure: Standard normal distribution with color-coded regions showing probability percentages

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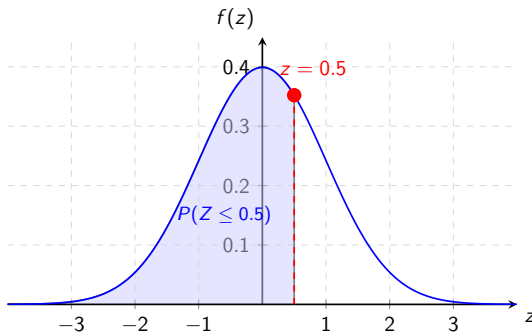


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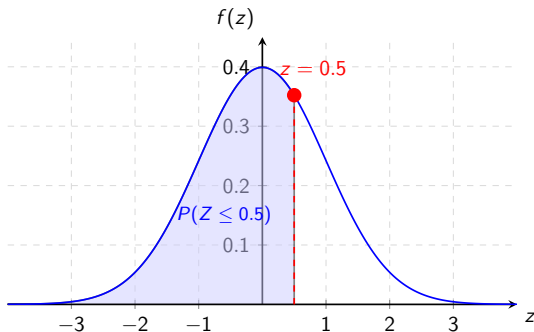


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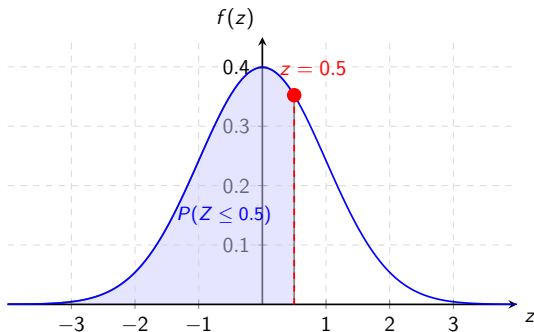


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- Thus, in the above figure: $p = P(Z \leq z_p) = \Phi(z_p)$

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Recognizing that the integrand is the PDF of a standard normal distribution, we have:

Probability of a normal random variable (cont.)

Given a normal r.v. $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$P(a < X \leq b) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \quad (8)$$

Substituting $z = \frac{x-\mu}{\sigma}$ and $dx = \sigma dz$, we obtain:

$$P(a < X \leq b) = \frac{1}{\sqrt{2\pi}} \int_{(a-\mu)/\sigma}^{(b-\mu)/\sigma} e^{-\frac{1}{2}z^2} dz \quad (9)$$

Recognizing that the integrand is the PDF of a standard normal distribution, we have:

$$P(a < X \leq b) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right) \quad (10)$$

Example 5a: Normal probabilities

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SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

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SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (a) What is the probability that a randomly selected student has a score that is at least 1200? The Z-score is $z = \frac{1200 - 1100}{200} = 0.5$

Example 5a: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (a) What is the probability that a randomly selected student has a score that is at least 1200? The Z-score is $z = \frac{1200-1100}{200} = 0.5$

Thus,

$$P(X \geq 1200) = 1 - \Phi(.5) =$$

Example 5a: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (a) What is the probability that a randomly selected student has a score that is at least 1200? The Z-score is $z = \frac{1200-1100}{200} = 0.5$

Thus,

$$P(X \geq 1200) = 1 - \Phi(.5) = 1 - .695 = \boxed{.3085}$$

Example 5a: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (a) What is the probability that a randomly selected student has a score that is at least 1200? The Z-score is $z = \frac{1200-1100}{200} = 0.5$

Thus,

$$P(X \geq 1200) = 1 - \Phi(.5) = 1 - .695 = .3085$$

In Python, you can compute this probability using the `scipy.stats` library:

```
import scipy.stats as stats
p = 1 - stats.norm.cdf(1200, 1100, 200)
```

The first 3 arguments of `stats.norm.cdf` are the value, mean (`textttloc`), and standard deviation (`textttscale`), respectively.

Example 5a: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

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```

Example 5a: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (a) What is the probability that a randomly selected student has a score that is at least 1200? The Z-score is $z = \frac{1200-1100}{200} = 0.5$

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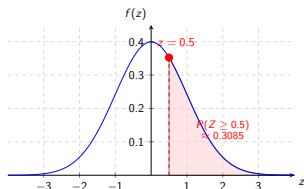
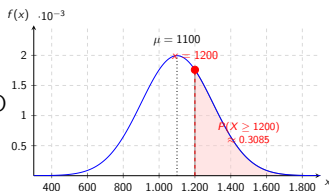
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The first 3 arguments of `stats.norm.cdf` are the value, mean (`textttloc`), and standard deviation (`textttscale`), respectively. OR, you can use the Z-score (default `mean=0`, `std=1`):

```
import scipy.stats as stats  
p = 1 - stats.norm.cdf(0.5)
```

which gives the same result



Example 5b: Normal probabilities

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SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

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SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (b) What is the probability that another randomly selected student's score is greater than 600 but less than 1200?

Example 5b: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (b) What is the probability that another randomly selected student's score is greater than 600 but less than 1200?

$$P(600 \leq X < 1200) = \Phi\left(\frac{1200 - 1100}{200}\right) - \Phi\left(\frac{600 - 1100}{200}\right)$$

Example 5b: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (b) What is the probability that another randomly selected student's score is greater than 600 but less than 1200?

$$\begin{aligned} P(600 \leq X < 1200) &= \Phi\left(\frac{1200 - 1100}{200}\right) - \Phi\left(\frac{600 - 1100}{200}\right) \\ &= \Phi(.5) - \Phi(-2.5) \end{aligned}$$

Example 5b: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (b) What is the probability that another randomly selected student's score is greater than 600 but less than 1200?

$$\begin{aligned} P(600 \leq X < 1200) &= \Phi\left(\frac{1200 - 1100}{200}\right) - \Phi\left(\frac{600 - 1100}{200}\right) \\ &= \Phi(.5) - \Phi(-2.5) = \boxed{.6853} \end{aligned}$$

In Python:

```
from scipy.stats import norm  
p = norm.cdf(1200, 1100, 200)  
    - norm.cdf(600, 1100, 200)
```

Example 5b: Normal probabilities

SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (b) What is the probability that another randomly selected student's score is greater than 600 but less than 1200?

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In Python:

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from scipy.stats import norm  
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```

Example 5b: Normal probabilities

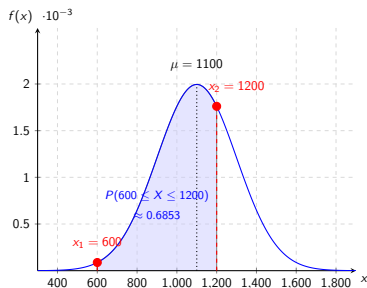
SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

- (b) What is the probability that another randomly selected student's score is greater than 600 but less than 1200?

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In Python:

```
from scipy.stats import norm  
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```



Example 5b: Normal probabilities (cont)

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(b) OR, you can use the Z-scores (default mean=0, std=1):

```
import scipy.stats as stats  
p = norm.cdf(0.5) - norm.cdf(-2.5)
```

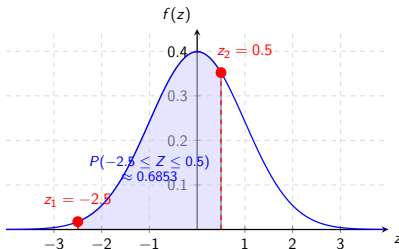


Figure: Standard normal PDF with probability area shaded

Example 5b: Normal probabilities (cont)

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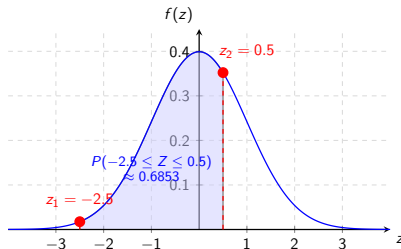


Figure: Standard normal PDF with probability area shaded

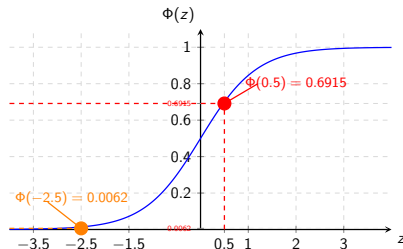


Figure: Standard normal CDF with corresponding probability values marked

Example 6: Inverse normal probabilities

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SAT scores are normally distributed as $X \sim \mathcal{N}(1100, \sigma = 200)$.

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$$\therefore x = z\sigma + \mu = -.2533(200) + 1100 = \boxed{1049}$$

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$$z = \Phi^{-1}(.4) = -.2533 = \frac{x - \mu}{\sigma}$$

$$\therefore x = z\sigma + \mu = -.2533(200) + 1100 = \boxed{1049}$$

```
from scipy.stats import norm  
p = norm.ppf(0.4, 1100, 200)
```

Example 6: Inverse normal probabilities

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from scipy.stats import norm  
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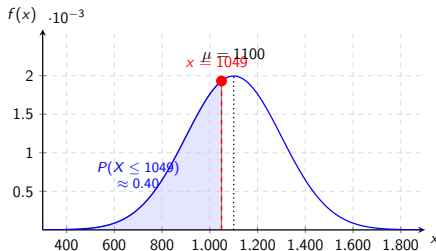


Figure: Normal distribution with $\mu = 1100$, $\sigma = 200$, and $P(X \leq 1049)$ shaded

Example 6: Inverse normal probabilities (cont.)

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- (c) You can think of the inverse CDF as finding the x value that corresponds to a given percentile.

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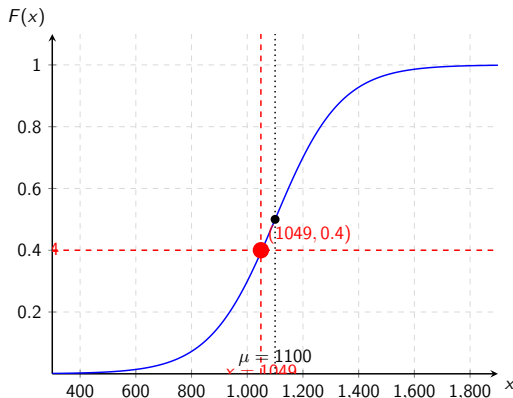


Figure: Normal distribution CDF showing the 40th percentile at $x = 1049$

More on the normal CDF

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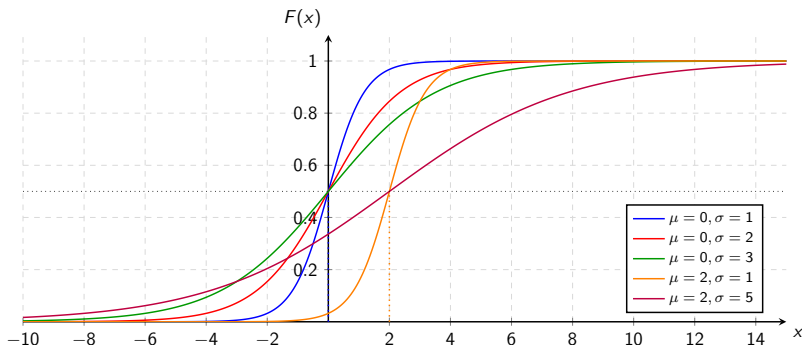


Figure: Comparison of normal distribution CDFs with different parameters

More on the normal CDF

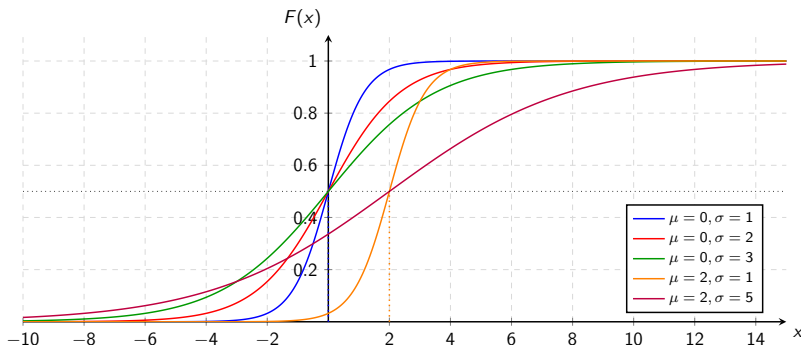


Figure: Comparison of normal distribution CDFs with different parameters

- The standard normal CDF is the blue curve in the above figure

More on the normal CDF

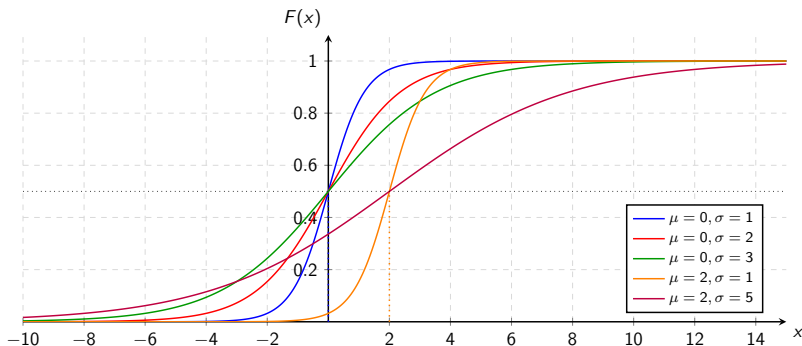


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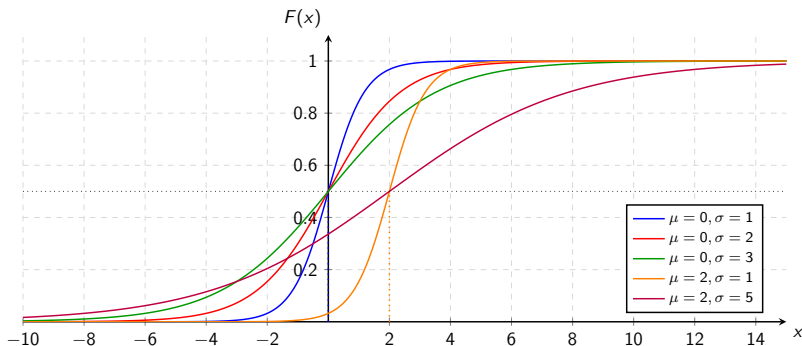


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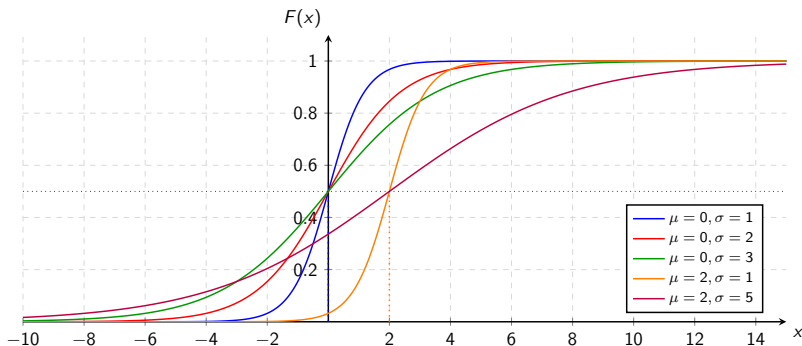


Figure: Comparison of normal distribution CDFs with different parameters

- The standard normal CDF is the blue curve in the above figure
- Quantiles can be read off the plot (e.g. the median is the value of X corresponding to the y value of 0.5)
- $\Phi(-z) = 1 - \Phi(z)$
- $z = \Phi^{-1}(p) = -\Phi^{-1}(1 - p)$

Example 7: Probability of flooding

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The drainage from a community during a storm is a normal random variable estimated to have a mean of 1.2 million gallons per day (mgd) and an SD of 0.4 mgd. If the storm drain system is designed with a maximum drainage capacity of 1.5 mgd:

Example 7: Probability of flooding

The drainage from a community during a storm is a normal random variable estimated to have a mean of 1.2 million gallons per day (mgd) and an SD of 0.4 mgd. If the storm drain system is designed with a maximum drainage capacity of 1.5 mgd:

- (a) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system?
- (b) Find $P(1.0 < X \leq 1.6)$.
- (c) Find the 90th-percentile drainage load from the community during a storm.

Example 7: Probability of flooding (cont.)

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Given $\mu = 1.2$ and $\sigma = 0.4$.

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Solution

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Given $\mu = 1.2$ and $\sigma = 0.4$.

- (a) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system?

Solution

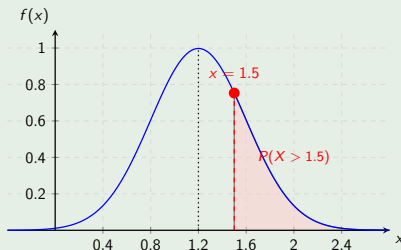


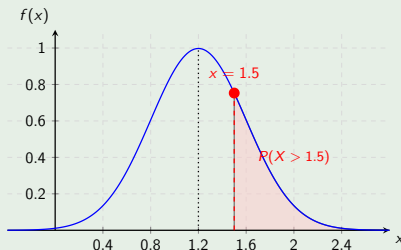
Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(X > 1.5)$ shaded

Example 7: Probability of flooding (cont.)

Given $\mu = 1.2$ and $\sigma = 0.4$.

- (a) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system?

Solution



$$P(X > 1.5) = 1 - P(X \leq 1.5)$$

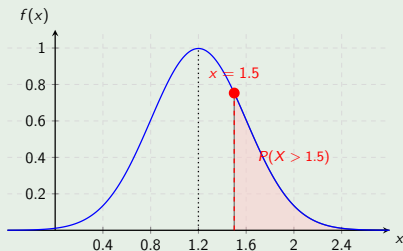
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Example 7: Probability of flooding (cont.)

Given $\mu = 1.2$ and $\sigma = 0.4$.

- (a) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system?

Solution



$$\begin{aligned} P(X > 1.5) &= 1 - P(X \leq 1.5) \\ &= 1 - \Phi\left(\frac{1.5 - 1.2}{0.4}\right) \end{aligned}$$

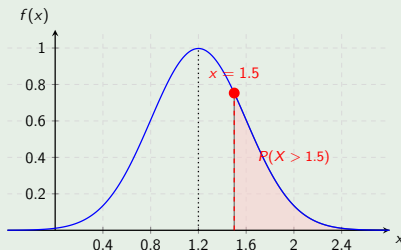
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Example 7: Probability of flooding (cont.)

Given $\mu = 1.2$ and $\sigma = 0.4$.

- (a) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system?

Solution



$$\begin{aligned} P(X > 1.5) &= 1 - P(X \leq 1.5) \\ &= 1 - \Phi\left(\frac{1.5 - 1.2}{0.4}\right) \\ &= 1 - \Phi(0.75) \end{aligned}$$

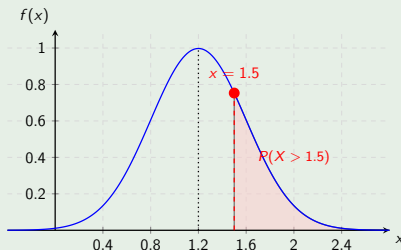
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Example 7: Probability of flooding (cont.)

Given $\mu = 1.2$ and $\sigma = 0.4$.

- (a) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system?

Solution



$$\begin{aligned} P(X > 1.5) &= 1 - P(X \leq 1.5) \\ &= 1 - \Phi\left(\frac{1.5 - 1.2}{0.4}\right) \\ &= 1 - \Phi(0.75) \\ &= 1 - 0.7734 = \boxed{0.227} \end{aligned}$$

Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(X > 1.5)$ shaded

Example 7: Probability of flooding (cont.)

Given $\mu = 1.2$ and $\sigma = 0.4$.

- (a) What is the underlying probability of flooding during a storm that is assumed in the design of the drainage system?

Solution

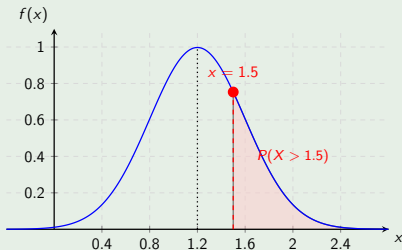


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In Python: `1 - norm.cdf(1.5, 1.2, 0.4)`

Example 7: Probability of flooding (cont.)

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(b) Find $p = P(1.0 < X \leq 1.6)$:

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Solution

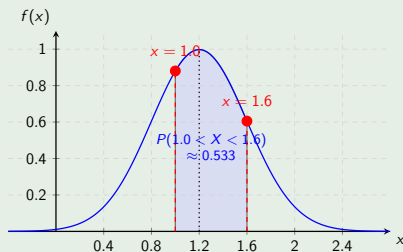
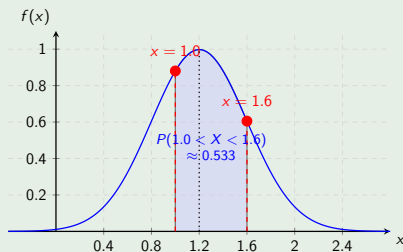


Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(1.0 < X \leq 1.6)$ shaded

Example 7: Probability of flooding (cont.)

(b) Find $p = P(1.0 < X \leq 1.6)$:

Solution



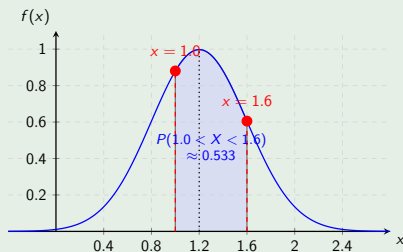
$$p = \Phi\left(\frac{1.6 - 1.2}{0.4}\right) - \Phi\left(\frac{1.0 - 1.2}{0.4}\right)$$

Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(1.0 < X \leq 1.6)$ shaded

Example 7: Probability of flooding (cont.)

(b) Find $p = P(1.0 < X \leq 1.6)$:

Solution



$$\begin{aligned} p &= \Phi\left(\frac{1.6 - 1.2}{0.4}\right) \\ &\quad - \Phi\left(\frac{1.0 - 1.2}{0.4}\right) \\ &= \Phi(1.0) - \Phi(-0.5) \end{aligned}$$

Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(1.0 < X \leq 1.6)$ shaded

Example 7: Probability of flooding (cont.)

(b) Find $p = P(1.0 < X \leq 1.6)$:

Solution

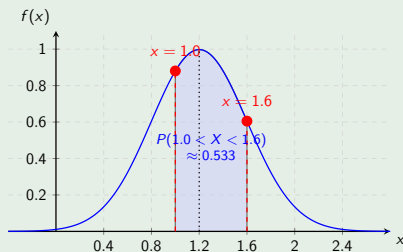


Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(1.0 < X < 1.6)$ shaded

$$\begin{aligned} p &= \Phi\left(\frac{1.6 - 1.2}{0.4}\right) \\ &\quad - \Phi\left(\frac{1.0 - 1.2}{0.4}\right) \\ &= \Phi(1.0) - \Phi(-0.5) \\ &= 0.8413 - [1 - \Phi(0.5)] \end{aligned}$$

Example 7: Probability of flooding (cont.)

(b) Find $p = P(1.0 < X \leq 1.6)$:

Solution

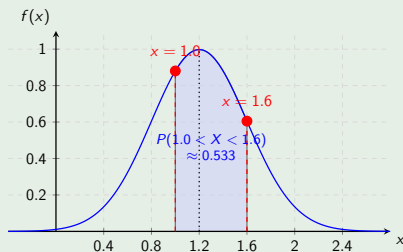


Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(1.0 < X < 1.6)$ shaded

$$\begin{aligned} p &= \Phi\left(\frac{1.6 - 1.2}{0.4}\right) \\ &\quad - \Phi\left(\frac{1.0 - 1.2}{0.4}\right) \\ &= \Phi(1.0) - \Phi(-0.5) \\ &= 0.8413 - [1 - \Phi(0.5)] \\ &= 0.8413 - (1 - 0.6915) \end{aligned}$$

Example 7: Probability of flooding (cont.)

(b) Find $p = P(1.0 < X \leq 1.6)$:

Solution

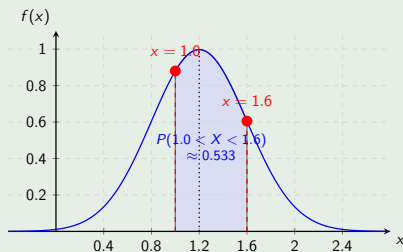


Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(1.0 < X \leq 1.6)$ shaded

$$\begin{aligned} p &= \Phi\left(\frac{1.6 - 1.2}{0.4}\right) \\ &\quad - \Phi\left(\frac{1.0 - 1.2}{0.4}\right) \\ &= \Phi(1.0) - \Phi(-0.5) \\ &= 0.8413 - [1 - \Phi(0.5)] \\ &= 0.8413 - (1 - 0.6915) \\ &= 0.8413 - 0.3085 \end{aligned}$$

Example 7: Probability of flooding (cont.)

(b) Find $p = P(1.0 < X \leq 1.6)$:

Solution

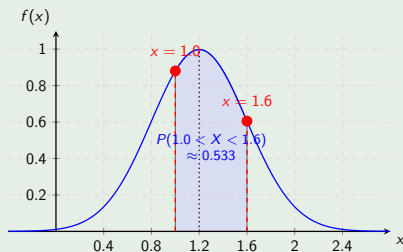


Figure: Normal distribution with $\mu = 1.2$, $\sigma = 0.4$, and $P(1.0 < X \leq 1.6)$ shaded

$$\begin{aligned} p &= \Phi\left(\frac{1.6 - 1.2}{0.4}\right) \\ &\quad - \Phi\left(\frac{1.0 - 1.2}{0.4}\right) \\ &= \Phi(1.0) - \Phi(-0.5) \\ &= 0.8413 - [1 - \Phi(0.5)] \\ &= 0.8413 - (1 - 0.6915) \\ &= 0.8413 - 0.3085 \\ &= 0.5328 \approx \boxed{0.533} \end{aligned}$$

Example 7: Probability of flooding (cont.)

(b) Find $p = P(1.0 < X \leq 1.6)$:

Solution

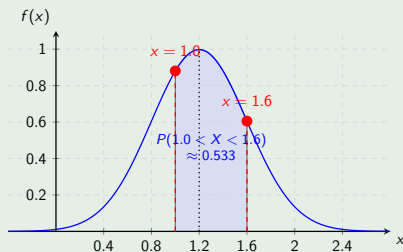


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In Python: `norm.cdf(1.6, 1.2, 0.4) - norm.cdf(1.0, 1.2, 0.4)`

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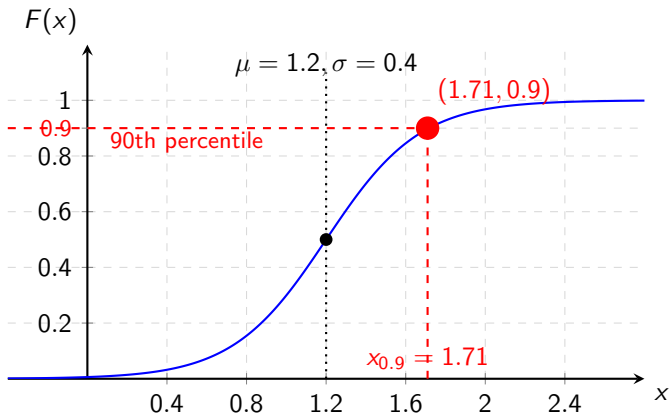


Figure: Normal CDF showing the 90th percentile at $x = 1.71$

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In Python:

```
from scipy.stats import norm  
p90 = norm.ppf(0.9, 1.2, 0.4)
```

gives 1.7095 mgd

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- The symbol Φ (“phi”) is used to represent the CDF of the *standard normal distribution*, whose values can be looked up in a table.
- In Python, the `scipy.stats.norm.cdf(x, mu, sigma)` and `scipy.stats.norm.ppf(p, mu, sigma)` can be used to compute probabilities and inverse CDFs of the normal distribution, respectively.